

Sibasish Barik

Indore, Madhya Pradesh | [LinkedIn](#) | +91 95565 97254 | ce230004044@iiti.ac.in

EDUCATION

Indian Institute of Technology Indore Indore, Madhya Pradesh
Bachelor of Technology in Civil Engineering , Final Year 2023 - 2027

SBD International School Bhadrak, Odisha
CBSE 10th and 12th 2014 - 2022

EXPERIENCE

Charak Center for Digital Healthcare, IIT Indore Indore, Madhya Pradesh
SIDDHI 3.0 Intern May 2025 - July 2025

- Developed and field-tested an autonomous drone prototype designed for medical supply delivery to remote areas. Successfully validated essential autonomous features, such as GPS waypoint navigation and precision landing, for a proposed 25km delivery system.

Center for System Design, NITK Surathkal Surathkal, Karnataka
Project Intern May 2026 - July 2026

- Developed a ROS 2 based simulation and hardware integration stack for an autonomous ground-effect aircraft using Gazebo and ArduPilot SITL. Programmed a reactive, vision-based obstacle avoidance node utilizing depth-to-laserscan translation and containerized the hardware deployment environment using Docker on Raspberry Pi 5.

FACULTY COLLABORATIVE PROJECTS

Light Detecting and Following Mini Drone for navigation in GPS denied environments IIT Indore
Dr.Sumit Gautam (Dept. of Electrical Engineering, SwaYaan Lab) April 2025 - February 2026

- Integration of PlutoX drone with ESP32 for light detection using TCS34725 RGB Sensor
- Development of advanced control algorithm for drone navigation using camera module

OpenMower- Autonomous Lawnmower with integrated robotic arm IIT Indore
Dr.Vimal Bhatia (Dept. of Electrical Engineering) August 2025 - March 2026

- HW/SW integration of sensors with Arduino and Raspberry pi

System and method for wirelessly charging nano-drones using tethered drone platform IIT Indore
Dr.Sumit Gautam (Dept. of Electrical Engineering, SwaYaan Lab) August 2025- Present

- Provisional patent

UAV-Based Water Quality Monitoring and Sampling System for Rapid Environmental Assessment IIT Indore
Dr.Mayur Shirish Jain, (Dept. of Civil Engineering, Environmental Engineering Lab) March 2026- Present

- Conducting a comprehensive literature review and proof-of-concept study for an autonomous UAV-based water quality monitoring and sampling system.
- Designing a tether-assisted sensing mechanism for integrating **pH, EC, TDS, temperature, and turbidity sensors** to enable geo-referenced, depth-specific measurements.
- Evaluating the system's commercial viability through **Techno-Economic Analysis (TEA)** incorporating **CAPEX/OPEX estimation, Net Present Value (NPV), Internal Rate of Return (IRR), Benefit-Cost Ratio, Payback Period, and sensitivity analysis.**

TECHNICAL PROJECTS

Autonomous Delivery Drone Nov. 2024- May 2025

- Developed autonomous drone using Raspberry pi and Pixhawk Flight Controller for package delivery over a distance of upto 5km.
- Established communication between companion computer and Pixhawk via Mavlink using MAVSDK-Python

Gesture controlled Drone July. 2025- Aug 2025

- Developed a hand gesture controlled drone using Raspberry pi and Pixhawk Flight Controller

Autonomous dual-drone system for rescue operation (NIDAR Challenge)

April 2025 - January 2026

- Developed a system for dual drone communication over long distances for transmission of coordinates of geotagged people.
- Developed algorithm for simple autonomous navigation of both surveying and delivery drones using python script.

Control system for Drone Stabilization and Precision Motion (Inter IIT Techmeet 14.0)

December 2025

- Designed and implemented a cascaded PID control architecture for drone stabilization and precision motion.
- Developed sensor-fusion and filtering algorithms using a Linear Kalman Filter and BiQuad filtering to reduce IMU and altitude sensor noise, enabling smoother velocity estimation and more stable flight behavior.
- Secured 8th position in Low-Prep PS among 23 IITs.

Electric Vehicle and Intelligent Ground Vehicle

April 2024 - August 2025

- Developed and integrated the core hardware and electronics systems for an Electric Vehicle and an Intelligent ground vehicle for autonomous obstacle avoidance and path planning.

Miscellaneous Projects

2019 - 2022

- Developed an ESP32-CAM home surveillance system that uses a proximity sensor to detect intruders and sends real-time images via Telegram API.
- Voice controlled and smartphone remote controlled robot car built using Arduino

Autonomous UAV Docking and Charging System (IROC-U 2026)

January 2026 - July 2026

- Designed the complete drone hardware architecture and led CAD design, hardware integration, prototyping, and system validation for autonomous docking operations.

TECHNICAL LEADERSHIP EXPERIENCE

Head, Aeromodelling Club, IIT Indore

Inventory Head, Tinkerers' Lab, IIT Indore

Team Lead, National Innovation for Drone Application and Research Challenge (NIDAR) 2025

Team Lead, NXP AIM 2024

POSITIONS OF RESPONSIBILITY

Member, Intelligent Vehicle Design and Control Club, IIT Indore

Member, Astronomy Club, IIT Indore

Member, Photography Club, IIT Indore

Member, Animal Welfare Team (Hobby Club), IIT Indore

Member, Concreate Club, IIT Indore

TECHNICAL SKILLS & CERTIFICATIONS

Programming: Python, C/C++, Arduino programming, HTML, CSS and Javascript basics

Hardware: Raspberry Pi, NVIDIA Jetson, Arduino, ESP32, Teensy, HW/SW integration, Sensor integration, Drone development(Pixhawk/Ardupilot)

Tools/Frameworks: OpenCV, Dronekit-Python, MAVSDK-Python, Pymavlink, Docker

CAD: Fusion 360, LaserCAD, LightBurn, Solidworks

Certifications: SOLIDWORKS Associate Course, SOLIDWORKS Professional Course, Aircraft Systems and Avionics, Aircraft Fundamentals, MATLAB Onramp

ACHIEVEMENTS

3rd Rank Overall, Inter IIT Techmeet 14.0

32nd Rank, NIDAR Challenge 2025

Among the **top 74 teams**, IROC-U 2026

Qualifier, Team lead in NXP AIM 2024

Gold Medal, IITI Summer of Code, Concreate Domain

Silver Medal, Inter IIT Civil Conclave

Gold Medal, Photography contingent member and Winner- T vs M